Can Multi-agent Collaboration Break the Curse of Complexity?

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Background

LLMs have shown limited performance across reasoning tasks when evaluated as individuals. Prior work by Yuchen et al. [1] highlights a 'curse of complexity' scaling law in the context of reasoning.

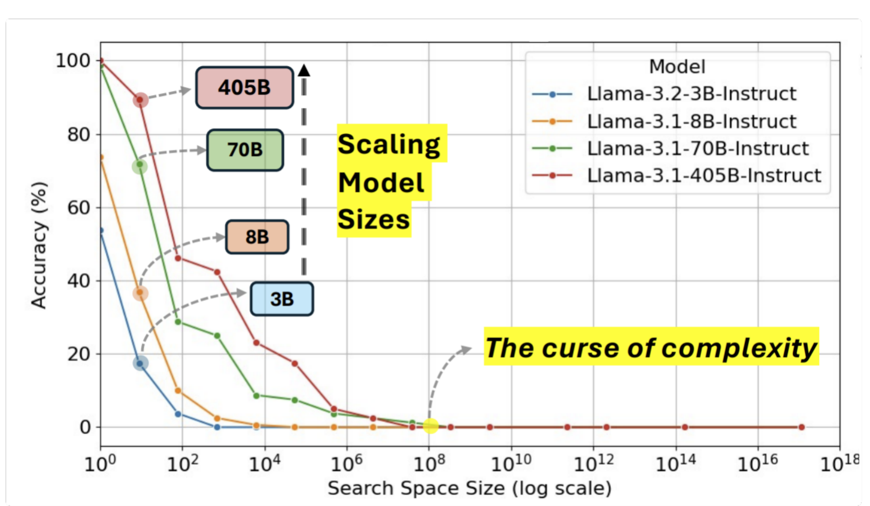

Figure 1:Scaling model size yields limited improvements on tasks higher in complexity [1]

Observation of human performance on logical reasoning tasks as individuals is similarly limited. [2] found that groups outperform the individuals that compose them on the Wason selection task. This phenomenon is sometimes referred to as the 'assembly bonus effect', wherein a group achieves better outcomes than its most competent individual.

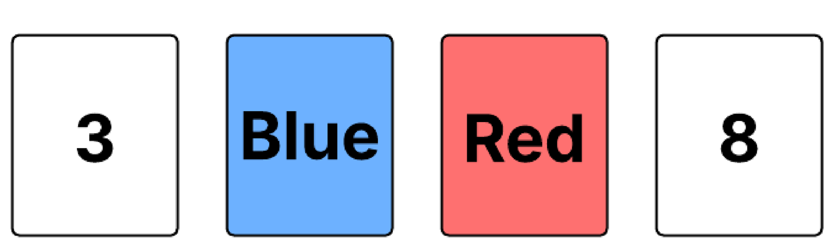

Figure 2:The Wason selection task

While the cognitive benefits of collaboration have been well-documented in human groups, it remains unclear whether similar effects can be observed in multi-agent systems composed of LLMs.

Research Questions

- Do collaborative LLM agents outperform individual models as task complexity increases?
- Can groups of LLM agents collectively outperform their strongest individual member on reasoning tasks?

Methods

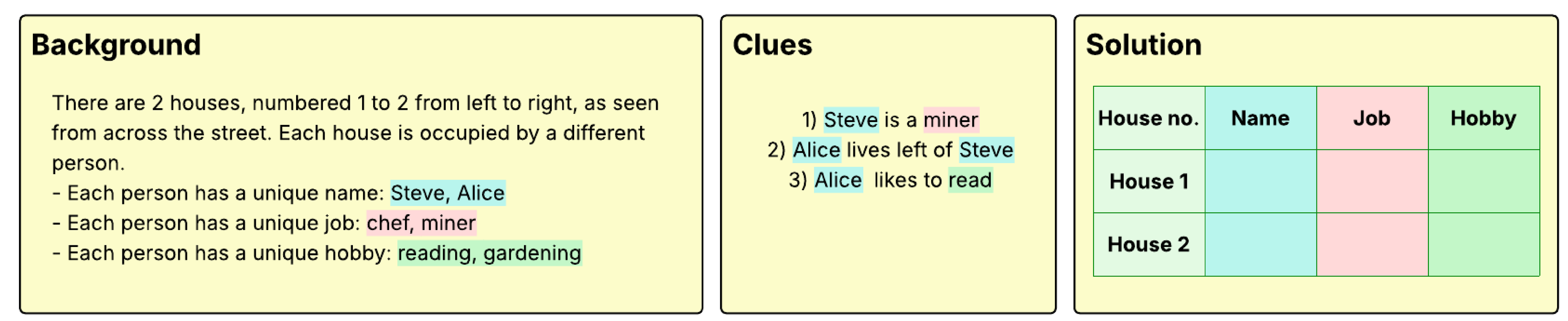

Figure 3:Example ZebraLogic puzzle

ZebraLogic benchmark consists of logic grid puzzles derived from constraint satisfaction problems (CSPs). Fig. 3 demonstrates a simple 2x3 problem. The task for the model is to determine the correct assignment of attributes to each house based on the clues given. While this example is easily solvable using linear deduction, as complexity grows, non-monotonic reasoning strategies are required to successfully solve the puzzles.

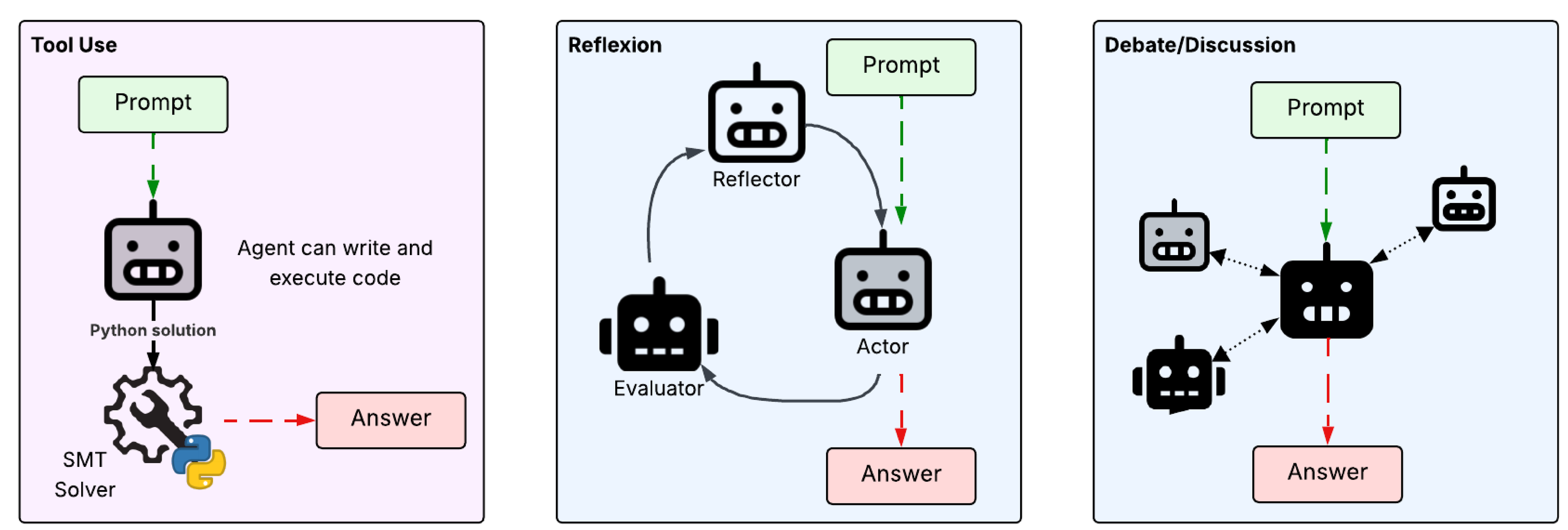

Figure 4:From left to right: a single agent approach with tool use, a multi-agent reflection approach, and a collaborative approach.

Complexity can be measured using the size of the search space, where an $N \times M$ grid has a solution space of size $(N!)^M$, with N representing the number of houses and M the number of attributes. The puzzles can be categorised into groups based on the magnitude of the search space $|S|$.
Z3 conflicts can also be used as a complementary measure of complexity. This is the average number of situations in which the popular satisfiability modulo theories (SMT) solver has to backtrack due to contradictions in its current assignment. Puzzles with higher conflicts require extensive backtracking, indicating higher logical complexity.

Metrics

Puzzle accuracy and cell-wise accuracy are measured to evaluate the completeness of the puzzle. The length of the reasoning section of the answer will also be observed to see if reasoning length and puzzle complexity are correlated.

Experiment

To investigate the research questions, various architectures of multi-agent (MA) systems, and single LLMs will be tested using the ZebraLogic benchmark described above. The MA systems should only be comprised of the LLMs used as controls for a fair comparison.

Preliminary Results

Model	Format	Puzzle Acc.	Cell Acc.
7B Math	JSON	15.63	25.63
	XML	2.19	1.91
7B Instruct	JSON	35.62	64.92
	XML	32.19	63.81
14B Code	JSON	43.44	70.32
	XML	34.69	69.13
14B Instruct	JSON	57.81	79.25
	XML	58.44	79.33

Table 1:Subset of results. All models are in the Qwen2.5 family.

An initial experiment evaluated the performance of LLMs on the ZebraLogic task when prompted to output solutions in either JSON or XML format.

Next Steps

Future work will apply the current experimental framework to a broader range of logical reasoning tasks to assess whether the observed patterns generalise across different problem types and domains. This will help evaluate the robustness of collaborative LLM agent performance and the potential applicability of the assembly bonus effect beyond the initial task setting.

References

[1] Bill Yuchen et al. Lin.
ZebraLogic: On the Scaling Limits of LLMs for Logical Reasoning, February 2025.

[2] David Moshman and Molly Geil.
Collaborative Reasoning: Evidence for Collective Rationality.
Thinking & Reasoning, 4(3):231–248, July 1998.

Acknowledgements

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